SOFTWARE

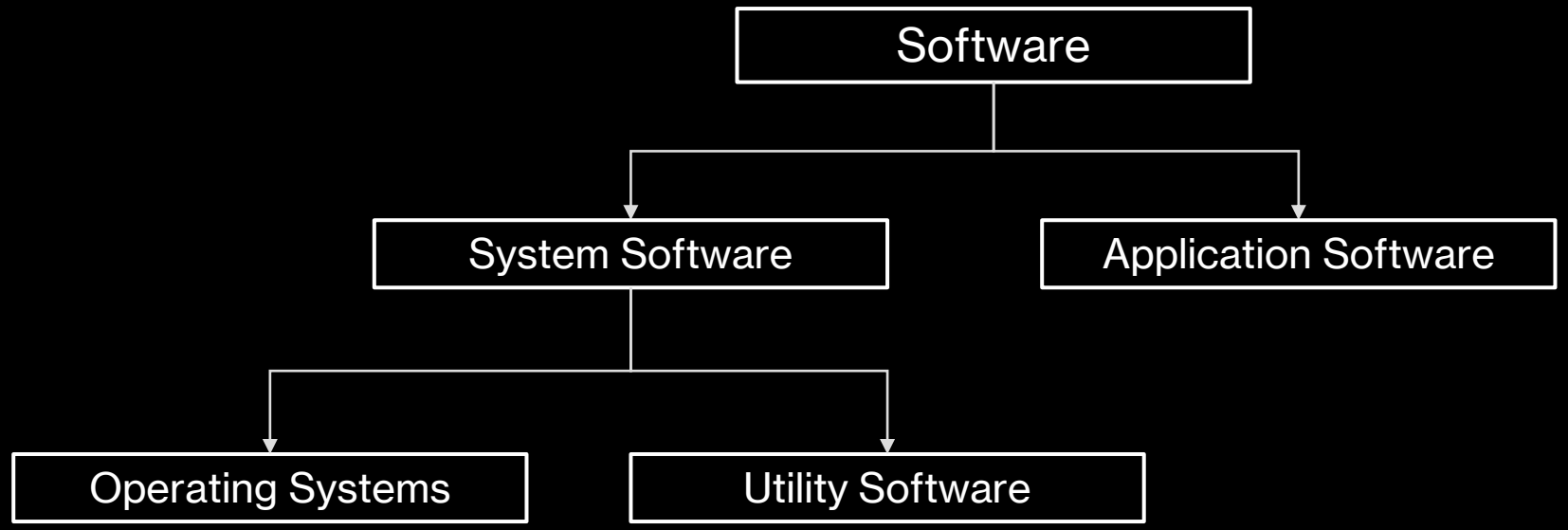
Week 6

LEARNING OBJECTIVES

1. Know what an operating system is and how it manages files, processes, hardware and the user interface
2. Understand the purpose and functions of utility software (managing, repairing and converting files; compression; defragmentation; backing up; antivirus, anti-spyware)
3. Understand how software can be used to simulate and model aspects of the real world
4. Understand what is meant by system software and application software

WHAT IS SOFTWARE?

Computer systems consist of both hardware and software. Software is the **set of programs** that run on a computer system.



SUBJECT VOCABULARY

- Application Software: software that performs a task that would otherwise be done by hand, perhaps with pen and paper
- System Software: operating system and utility software
- Operating System: software designed for hardware, and which manages other programs' access to the hardware
- Utility Software: software that does a useful job for the user that is not essential to the operating system and not the reason for using a computer in the first place

Application Programming Interface (API)

- **Code that allows two programs to communicate with each other.**

Using the operating system API rather than accessing hardware directly means that an application program will run on the operating system. It doesn't matter what hardware it is installed on. This helps to reduce the risk of programs crashing and to maintain security.

Paging & Scheduling

- **Scheduling:** the algorithm that the OS uses to allow each running process to use the CPU.
- **Paging:** the algorithm the OS uses to move programs from RAM to disk and back again when needed once main memory is full.

More on the Operating Systems

As well as the applications you are running, the operating system is also running other programs in the background. These programs are running one at a time, being given their share of the CPU time by the scheduler. They are all running alongside one another, but not really at the same time. Therefore:

- **Concurrent**: processes that run at the same time are described as being concurrent

Cont.

- Authentication: the process of proving to a computer system who you are (e.g. using a username and password).
- User interface the way the user interacts with the operating system.
  - Graphical User Interface (GUI): with windows, icons, menus and pointers like Microsoft Windows or MacOS.
  - Command-line Interface (CLI): Microsoft DOS.

Check pages (187-189)